Tensor Polarimetry with One and Two L/R/U/D Stations

DPOLAR analysis note

1 Why a tensor-only description is appropriate

The magnetic transport of a deuteron is governed, to leading order, by a Hamiltonian linear in the spin,

$$H(t) = -\boldsymbol{\mu} \cdot \boldsymbol{B}(t) \propto \boldsymbol{B}(t) \cdot \boldsymbol{S}. \tag{1}$$

The corresponding evolution operator is a spin rotation. Under such a rotation, irreducible spin tensors of different rank do not mix:

$$U(R)\, \tau_q^{(k)}\, U^\dagger(R) = \sum_{q'} D_{q'q}^{(k)}(R) \tau_{q'}^{(k)}. \tag{2}$$

Vector polarization is rank one and tensor polarization is rank two. Equation (2) therefore implies that a purely tensor-polarized beam remains purely tensor polarized during coherent magnetic transport. The tensor rotates, but it does not generate a vector polarization.

This separation remains true if different particles experience different magnetic rotations, provided the trajectory weights are spin independent: averaging rotation matrices within the beam ensemble still acts separately in each irreducible rank. Rank mixing would require physics beyond Eq. (1), such as a relevant interaction quadratic in spin or spin-selective loss. Neither is part of the present beam-transport model. It is therefore natural to restrict the polarimeter analysis to the five tensor degrees of freedom.

In Cartesian form,

$$\boldsymbol{P} = \begin{pmatrix} p_{xx} & p_{xy} & p_{xz} \\ p_{xy} & p_{yy} & p_{yz} \\ p_{xz} & p_{yz} & p_{zz} \end{pmatrix}, \qquad p_{xx} + p_{yy} + p_{zz} = 0. \tag{3}$$

A convenient physical coordinate set is

$$\boldsymbol{r} = (p_{xx} - p_{yy},\ p_{xy},\ p_{xz},\ p_{yz},\ p_{zz})^\mathsf{T}. \tag{4}$$

2 Information from one four-arm station

With vector polarization omitted, the polarized d–p cross section is

$$\begin{aligned}\frac{\sigma(\theta,\phi)}{\sigma_0(\theta)} = 1 &+ \frac{2}{3}(p_{xz}\cos\phi - p_{yz}\sin\phi)A_{xz}(\theta) \\ &+ \frac{1}{6}\left[(p_{xx} - p_{yy})\cos 2\phi - 2p_{xy}\sin 2\phi\right]\left[A_{xx}(\theta) - A_{yy}(\theta)\right] \\ &+ \frac{1}{2}p_{zz}A_{zz}(\theta).\end{aligned} \tag{5}$$

For ideal arms at $\phi = (0°, 90°, 180°, 270°)$, define $B = A_{xx} - A_{yy}$. Apart from the unpolarized rate and a common acceptance factor, the tensor response is

$$H_{\text{ring}} = \begin{pmatrix} B/6 & 0 & 2A_{xz}/3 & 0 & A_{zz}/2 \\ -B/6 & 0 & 0 & -2A_{xz}/3 & A_{zz}/2 \\ B/6 & 0 & -2A_{xz}/3 & 0 & A_{zz}/2 \\ -B/6 & 0 & 0 & 2A_{xz}/3 & A_{zz}/2 \end{pmatrix}. \tag{6}$$

The measured combinations are transparent:

$$L - R \propto p_{xz} A_{xz}, \tag{7}$$

$$U - D \propto -p_{yz} A_{xz}, \tag{8}$$

$$(L + R) - (U + D) \propto (p_{xx} - p_{yy}) B. \tag{9}$$

The component p_{xy} is absent because $\sin 2\phi = 0$ at all four cardinal azimuths. It is an exact geometrical null direction, not a weak sensitivity. Symmetric finite detector openings do not change this conclusion.

The p_{zz} term is common to all four arms. At a single polar angle it is indistinguishable from luminosity if the absolute normalization is free. The rank is then three: one diagonal anisotropy and two off-diagonal components are measured. If the absolute normalization is known, p_{zz} supplies a fourth independent observable.

A practical station contains more than one polar-angle group. With one shared luminosity, p_{zz} is separated from normalization when the acceptance-averaged A_{zz} values differ between angular groups. The station then determines

$$p_{xx} - p_{yy}, \qquad p_{xz}, \qquad p_{yz}, \qquad p_{zz}, \tag{10}$$

and has rank four out of five. The remaining null is exactly p_{xy}. If each angle has an independent unconstrained normalization, the p_{zz} information is lost and the effective rank returns to three.

For the planned two-angle proton geometry, the numerical response calculation gives rank 4/5 after profiling one shared luminosity. Its null singular vector is pure p_{xy} to numerical precision.

3 Information from two stations

Let $\boldsymbol{q}_0$ denote the tensor at a common reference location. Spin transport to station s gives

$$\boldsymbol{P}_s = R_s \boldsymbol{P}_0 R_s^{\mathsf{T}}, \qquad \boldsymbol{q}_s = T^{(2)}(R_s) \boldsymbol{q}_0. \tag{11}$$

The combined response of two stations is

$$J_{\text{combined}} = \begin{pmatrix} J_1 T^{(2)}(R_1) \\ J_2 T^{(2)}(R_2) \end{pmatrix}, \tag{12}$$

and the unobservable tensor subspace is

$$\boxed{\mathcal{N}_{\text{combined}} = \ker[J_1 T^{(2)}(R_1)] \cap \ker[J_2 T^{(2)}(R_2)]}. \tag{13}$$

This equation contains the whole argument for a second polarimeter.

- If the stations have the same response and the transport rotations are aligned, both have the same p_{xy} null. The combined rank remains four. The second station improves statistical precision but determines no new tensor component.
- If the relative spin rotation maps the first station's p_{xy} null into $p_{xx} - p_{yy}$, p_{xz}, p_{yz}, or p_{zz} at the second station, the intersection in Eq. (13) is empty. The combined tensor response has full rank five.
- Special rotations can leave the null spaces aligned. For example, under a rotation by α about the beam axis,

  $$p'_{xy} = \frac{1}{2} \sin 2\alpha \, (p_{xx} - p_{yy}) + \cos 2\alpha \, p_{xy}. \tag{14}$$

  Generic angles mix p_{xy} into the visible diagonal anisotropy. At $\alpha = 0^\circ, 90^\circ, 180^\circ$, the p_{xy} direction maps back onto itself and no rank is gained.

The repository calculation shows the same behavior. Two identical stations with identity transport remain rank four. Generic relative rotations about x, y, or z give rank five, except at symmetry angles where the transported nulls realign. These rotation scans establish the mathematical condition; an actual beam-line conclusion requires the real spin-transfer map.

4 Different trajectories through the beam line

The rotation R_s is an idealization. A particle with phase-space coordinate ξ follows its own orbit and sees a rotation $R_s(\xi)$. The tensor delivered to the station is the ensemble average

$$\overline{\boldsymbol{P}}_s = \int d\xi\, f(\xi) R_s(\xi) \boldsymbol{P}_0 R_s^{\mathsf{T}}(\xi) = \overline{T}_s^{(2)} \boldsymbol{q}_0. \tag{15}$$

The averaged map $\overline{T}_s^{(2)}$ acts only within tensor space, so the tensor-only description remains closed. It is not generally an orthogonal rotation, however. A spread of spin phases can reduce the tensor norm and attenuate different tensor harmonics by different amounts.

For a Gaussian spread σ_ψ of rotations about the beam axis,

$$\langle e^{im\psi} \rangle = e^{im\langle\psi\rangle} e^{-m^2\sigma_\psi^2/2}. \tag{16}$$

The $m = 1$ tensor components (p_{xz}, p_{yz}) are reduced by $e^{-\sigma_\psi^2/2}$, while the $m = 2$ pair $(p_{xx} - p_{yy}, p_{xy})$ is reduced by $e^{-2\sigma_\psi^2}$. The component p_{zz} is unchanged by a rotation about z. An observed loss of tensor polarization can therefore result from phase averaging even when every particle undergoes coherent spin motion.

The detector acceptance adds a second effect. Beam position, angle, and momentum change the local scattering angles, and the accepted trajectories need not have the same spin-phase distribution in L, R, U, and D. The correct channel mean is then

$$\mu_k = \int d\xi\, f(\xi) w_k(\xi)\, \sigma_k \left[R_s(\xi) \boldsymbol{P}_0 R_s^{\mathsf{T}}(\xi) \right], \tag{17}$$

where $w_k(\xi)$ contains orbit acceptance and detector efficiency. A channel-dependent phase-space weight can imitate a tensor asymmetry or weaken the nominal rank-five complementarity of two stations.

A quantitative two-station analysis therefore needs the orbit and spin transfer map over the measured beam phase space, beam-position and angle information at both stations, and finite-acceptance detector response. Without these inputs, two polarimeters constrain only an effective averaged tensor; they do not separate the initial tensor from an arbitrary distribution of spin phases.

5 Conclusion

For magnetic transport generated by $\boldsymbol{B} \cdot \boldsymbol{S}$, vector and tensor polarizations evolve independently. A tensor-only analysis is therefore a closed and physically appropriate description of the present beam.

One multi-angle cardinal L/R/U/D station determines four of the five tensor components when a common normalization is controlled. Its exact null direction is p_{xy}. A second aligned station improves precision but not rank. A second station after a known complementary spin rotation can remove the p_{xy} null and determine the complete tensor polarization. Path-dependent spin rotations do not create vector polarization, but their ensemble average can attenuate the tensor and introduce additional transport and acceptance uncertainties. The case for two stations is consequently a statement about the transported null-space intersection and the beam's phase-space-dependent spin map.